THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 13: **Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ozone layer is present in the: 1

(a) stratosphere **(b)** exosphere **(c)** troposphere **(d)** mesosphere

2. The process by which water vapour changes to liquid in the atmosphere is: 1

(a) evaporation **(b)** melting **(c)** condensation **(d)** freezing

3. The outer core of the Earth is: 1

(a) liquid **(b)** solid **(c)** plasma **(d)** gas

4. The ozone layer protects us from: 1

(a) meteors only **(b)** infrared rays only

(c) harmful ultraviolet rays of the Sun **(d)** rain

5. Earth's magnetic field protects life from: 1

(a) harmful charged particles of the solar wind

(b) earthquakes

(c) meteors only

(d) rain

6. The inner core of the Earth is made mainly of: 1

(a) water **(b)** iron and nickel **(c)** granite **(d)** sand

7. Temperature in the troposphere with increasing height: 1

(a) increases **(b)** decreases

(c) stays same **(d)** becomes zero at once

8. Earth is at the right distance from the Sun so that water can exist as a: 1

(a) solid only **(b)** plasma **(c)** liquid **(d)** gas only

9. The envelope of gases surrounding the Earth is the: 1

(a) atmosphere **(b)** lithosphere **(c)** biosphere **(d)** hydrosphere

10. The layer of the atmosphere where most meteors burn is the: 1

(a) mesosphere **(b)** troposphere **(c)** crust **(d)** stratosphere

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the hydrosphere? Give two parts of it. 2

12. Name two sources of fresh water. 2

13. What is the importance of the ozone layer? 2

14. What is the lithosphere? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the temperature in the troposphere decreases with height. 3

16. Differentiate between the biosphere and the hydrosphere. 3

17. State three effects of air pollution on the atmosphere and the biosphere. 3

OR

Explain the importance of the ozone layer and how it is damaged. 3

18. List three ways of conserving water. 3

OR

Describe how the movement of water in the hydrosphere helps living things. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the four spheres of the Earth and how they interact with each other with examples. 5

OR

Explain the role of the atmosphere in supporting life with five points. 5

20. Explain why the Earth supports life, comparing it with Venus and Mars. 5

OR

Explain the origin and age of the Earth, the formation of its layers and the formation of the atmosphere and oceans in simple terms. 5

END OF QUESTION PAPER — Total: 40 Marks

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